

Analytics Project Portfolio

Supporting Project Evidence | Chirag Gupta | MS in Business Analytics

Purpose	Concise documentation of analytics projects listed on the resume, including public-safety, predictive modeling, geospatial analysis, and operational analytics.
Core Tools	Python, SQL, R, Power BI, XGBoost, K-Means, CartoDB, Folium/Leaflet, ggmap, time series analysis, simulation

1. Urban Safety Spatial Analysis

Mar 2026 | Philadelphia City Hall Area | Public Safety / Geospatial Analytics

- Analyzed spatial-temporal crime patterns near Philadelphia City Hall using SQL and the Philadelphia CARTO data source.
- Filtered selected theft and burglary incidents within an approximately 500-meter radius using a spatial proximity query and a six-month analysis window.
- Used Python, Pandas, Shapely, Folium, and SQL/CARTO to process incident data, extract time features, and map incident locations.
- Created an interactive incident map, an hourly crime-frequency chart, and a day-versus-hour heatmap to identify temporal patterns.
- Executed notebook results identified 17 selected incidents, with Sunday as the highest-frequency day, 22:00 as the peak hour, and Theft from Vehicle as the most common selected category.
- Resume application: proposed data-driven high-visibility patrol strategies based on the observed spatial-temporal patterns.

2. Ballpark Sales & Operational Optimization

Oct - Dec 2025 | Predictive Modeling / Operations Analytics

- Built an XGBoost predictive model to identify key revenue drivers, with model performance reported on the resume at approximately $R^2 = 0.85$.
- Applied K-Means clustering to segment operational patterns and support staffing strategy development.
- Combined predictive modeling and clustering to translate analytical findings into operational recommendations.

3. Crime Pattern Visualization & Safety Analytics

Jan - Mar 2025 | Geospatial / Time Series / Public Safety Analytics

- Developed geospatial maps and predictive analyses focused on crime-pattern identification.
- Used R, ggmap, Leaflet, and time series analysis to visualize location-based activity and identify peak crime trends.
- Demonstrated the ability to combine mapping, temporal analysis, and analytical interpretation in a public-safety context.

4. AI-Driven Procurement & Supply Chain Analytics

Sep - Dec 2024 | Business Intelligence / Risk Analytics

- Built a Power BI supplier KPI dashboard to support procurement performance monitoring and decision-making.
- Developed a vendor risk-scoring model to evaluate supplier risk.
- Used simulation to assess disruption scenarios; the resume reports an estimated 20%-25% reduction in disruption risk.

Capabilities Demonstrated Across Projects

Data & Programming	Python, SQL, R, Pandas, data transformation, querying, validation, feature creation
Statistical / Predictive	XGBoost, K-Means, time series analysis, predictive modeling, simulation
Visualization	Power BI, Folium, Leaflet, ggmap, spatial maps, charts, heatmaps, dashboards
Applied Domains	Public safety, revenue optimization, operations, procurement, supply-chain risk

Relevance to Research & Information Analysis

These projects demonstrate hands-on analytical work involving data querying, statistical and predictive methods, geospatial analysis, dashboards, time-based pattern analysis, and interpretation of findings. The two crime-focused projects are directly related to public-safety analytics, while the other projects demonstrate transferable research, modeling, operational, and decision-support skills.

Note: This portfolio summarizes academic/project work listed on the resume and is intended as supporting evidence of analytical capability. It is not presented as employment verification.